

SHAGBAOR HYCENT AMOOL

Curriculum Vitae

Bio-innovative Water and Wastewater Biorefinery Lab, School of Engineering, University of Northern British Columbia, Prince George, BC, Canada

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RESEARCH INTERESTS

Aerobic granular sludge and biological wastewater treatment; extracellular polymeric substances and biopolymer recovery (curdlan, EPS); resource recovery, wastewater biorefineries and the circular bioeconomy; electrocatalytic treatment and carbon utilization; microplastics in water; life cycle and environmental impact assessment.

EDUCATION

PhD, Environmental Engineering | University of Northern British Columbia, Canada **Sep 2026 – 2030 (exp.)**

Thesis: *Transforming brewery wastewater into a biopolymer biorefinery — curdlan production and recovery in aerobic granular sludge systems*. Supervisor: Dr. Oliver T. Iorhemen.

MSc, Global Sustainability Engineering — Distinction | Heriot-Watt University, United Kingdom **2024 – 2025**

Dissertation: *Integration of Advanced Electrocatalytic Wastewater Treatment with Carbon Utilisation*. Supervisors: Dr. Jeannie Ziang Yie Tan; Prof. John Andresen. Research Centre for Carbon Solutions.

BEng, Agricultural and Environmental Engineering — First Class | University of Agriculture Makurdi, Nigeria (now Joseph Sarwuan Tarka University Makurdi) **2013 – 2019**

Project: *Development of a Motorised Hand-Held Rice Harvester*. Supervisor: Prof. Theresa K. Kaankuka.

RESEARCH POSITIONS

Doctoral Researcher | Bio-innovative Water and Wastewater Biorefinery Lab, UNBC **Sep 2026 – Present**

- Aerobic granular sludge treating real brewery wastewater in bench-scale sequencing-batch reactors: granulation and water-quality monitoring (COD, ammonia, nitrite, nitrate, phosphorus); statistically designed optimization of operating parameters; algal-bacterial granulation; quorum-sensing regulation of EPS; curdlan extraction, quantification and benchmarking.

MSc Researcher | Research Centre for Carbon Solutions, Heriot-Watt University **Sep 2024 – May 2025**

- Synthesized five palladium nanocatalysts and evaluated electrocatalytic degradation of N-nitrosodiethanolamine, a carcinogenic carbon-capture by-product: 78.9% removal against a ~39% conventional benchmark, with chromatography confirming breakdown to monoethanolamine and diethanolamine.

Research Intern, Irrigation and Machinery | National Centre for Agricultural Mechanization, Nigeria **Oct 2016 – Mar 2017**

- Automatic weather station data management; irrigation scheduling (surface, sprinkler and drip); performance evaluation of agricultural machinery.

PUBLICATIONS

Peer-reviewed journal articles

- Lumun, T.C., Ikoye, B., **Amool, S.H.**, Shiaoondo, T.L., Barakur, C.A., Guusu, M.J., et al. (2025). Energy consumption patterns and conservation opportunities in Nigerian universities: case study of an energy audit of Benue State University, Makurdi. *Journal of Advance Research in Food, Agriculture and Environmental Science*. doi:10.61841/rmh2fm43

Manuscripts under review

- Amool, S.H.**, Vesuwe, R.N., Thring, R.W., Iorhemen, O.T. Transforming brewery wastewater treatment into a biopolymer biorefinery: optimizing curdlan biosynthesis and sustainable recovery in aerobic granular sludge systems. *Resources* (submitted Sep 2026; under review).
- Amool, S.H.**, Kaboggoza, H.C., Manga, M., Thring, R.W., Iorhemen, O.T. Aerobic granular sludge: an emerging next-generation biotechnology for tackling microplastic pollution. *Environmental Management: Engineering and Technology* (submitted Aug 2026; revision under preparation).

Preprints

- Amool, S.H.**, Tan, J.Z.Y., Andresen, J. (2026). Integration of advanced electrocatalytic wastewater treatment with carbon utilisation. *SSRN*. doi:10.2139/ssrn.6717086
- Mzough, O.M., Nyityo, A.I., Ijir, A., Lumun, T.C., Achimba, O., Terwase, A.I., ... **Amool, S.H.** (2026). A fractional-order reduced model for predicting long-term thermal recovery from depleted hydrocarbon reservoirs. *SSRN*. doi:10.2139/ssrn.7488653

CONFERENCE PRESENTATIONS

- Amool, S.H.** (2026). Electrochemical industrial symbiosis: turning carbon and wastewater into shared value. ECR Net Zero Conference, Birmingham, UK, 9–10 March (poster) — **Delegates' Prize, voted best of 33 posters.**

2. **Amool, S.H.** (2025). Integrated electrocatalytic wastewater treatment and carbon utilisation using Pd-based nanocatalysts. IDRIC Annual Conference, London, UK, 11–12 June (poster).
3. **Amool, S.H.** (2023). Trending climate-smart renewable energy technologies in Nigeria. *Proceedings of the Nigerian Institution of Agricultural Engineers*, 43, 839–852 (oral).
4. **Amool, S.H.**, Agidzi, D. (2023). Trending mulching technologies and benefits in soil and water conservation: a discourse for adoption in Nigeria. 4th Symposium, ISTRO Nigeria Chapter, Abuja (poster).
5. **Amool, S.H.**, Agidzi, D. (2023). X-raying the impact of tillage practices on properties of soils in Nigeria. 4th Symposium, ISTRO Nigeria Chapter, Abuja (oral).
6. Kaankuka, T.K., Egbe-Okpenge, I.S., **Amool, S.H.** (2019). Development of a motorised hand-held rice harvester. 40th Annual and 20th International Conference, Nigerian Institution of Engineers, 2–6 December (oral).

INVITED TALKS & RESEARCH COMMUNICATION

- **Amool, S.H.** (2026). From waste to worth: sustainable waste management, circular economy and green innovation; and Building green communities: environmental leadership and sustainable livelihoods. Invited two-part webinar series as Mentor and Lead Facilitator, Women Ascension One Million Leadership Fellowship, 1–2 October (online).
- Founder and convenor, *Via Scholaris* (2024–present): an open academic-development platform of recorded workshops on research proposals, statements of purpose, postgraduate funding and reference management — 89 videos, 1,000+ subscribers, with invited speakers from Bloomberg, Arup UK, ENODA and OCU Group.

TEACHING & SUPERVISION

Lecturer I | Department of Agricultural Technology, Federal Polytechnic Wannune, Nigeria **May 2022 – May 2026**

- Lectured Agricultural Technology, Sustainable Design and Engineering Materials, including research-methods and statistics content, across full cohorts of National and Higher National Diploma students.
- Supervised final-year student projects from problem definition through experimental work and technical analysis to oral defence.
- Set, marked and moderated continuous assessments and examinations; mentored students on project writing and presentation.

RESEARCH SKILLS

Laboratory	Bioreactor operation and monitoring; standard water-quality methods (COD, ammonia, nitrite, nitrate, phosphorus); nanocatalyst synthesis and electrocatalytic testing; chromatographic product analysis; EPS and biopolymer extraction with gravimetric and enzymatic assays
Quantitative	Design of experiments (Plackett–Burman, central composite); ANOVA, linear models, GLM, GAM and mixed models; PCA, NMDS and RDA; life cycle assessment; environmental impact assessment
Software	Excel, SPSS, Minitab, MATLAB, Origin, PVsyst; EndNote and Mendeley
Scholarly	Systematic literature synthesis; first-author manuscript drafting and revision; grant and proposal writing; poster and conference presentation

AWARDS & SCHOLARSHIPS

- **UNBC Doctoral Tuition Award**, University of Northern British Columbia (2026–2028).
- **Delegates' Prize**, ECR Net Zero Conference, Birmingham, UK (2026) — poster voted best of 33.
- **PTDF Overseas Scholarship**, Petroleum Technology Development Fund, Nigeria (2023/2024) — full funding for MSc study in the United Kingdom.
- **Best Graduating Student**, Department of Agricultural and Environmental Engineering and College of Engineering, University of Agriculture Makurdi (2021).
- Nominee, Principal's Award for Outstanding Contribution to the Student Community, Heriot-Watt University (2025); Nominee, Scottish Borders Campus Champion Award (2026).

ACADEMIC SERVICE & UNIVERSITY LEADERSHIP

- **Student Member of Parliament**, Heriot-Watt University Student Union (2024–2025).
- **Vice President**, Nigerians & Friends Society, Heriot-Watt University (2024–2025) — supported 200+ international students and coordinated three orientation programmes.
- **Social Media Coordinator**, NetZero Society; **Social Media Ambassador**, School of Engineering and Physical Sciences, Heriot-Watt University (2024–2025).

REFERENCES

Available on request.